

**Year 10 Science
Physical Sciences
Safety Features of Cars**

Over the years cars have become safer due to increased use of safety features, both inside and outside the vehicle. Some safety features have been around for over 50 years, but some are very recent, having been developed only in the last 10 - 20 years.

Listed below are some of the major safety features found in cars today. You may find others.

- Anti-lock Braking System (ABS).
- Air bags - driver, passenger, side curtain.
- Crumple zones.
- Inertia reel seat belts.
- Collapsible steering column.

Your task is to research **one** of these features, trace its history and development (improvements), and the Physics principles behind its working.

You must include the following information, plus any other interesting facts.

- Is it a primary or secondary safety feature? (Explain)
- When was it developed, by whom and why?
- How does it work /what does it do (including how it protects the driver/passenger/pedestrian)?
- Which Physics laws or principles are involved in the safety features?
- Early examples of the feature.
- Modern or recent improvements.

Presentation

You may present your research in any of the following forms.

- Booklet - A4 sheets stapled together.
- Poster - may have more than one sheet.
- Keynote presentation.
- Pages, Word, Notability file.

Criteria

- You will work **individually** - it is not a group project.
- Do not plagiarise. Read all the material you come across and put it into your own words.
- Technical and scientific terms that you don't know should be explained.
- At the end of the booklet put a reference list in an acceptable format.

e.g. Cook, A.J. 2001, *Safety Features of Modern Cars*, Allen and Unwin, Melbourne, pp.216-224.

Einstein, A. 1905, *How Physics Can Be Applied to the Model T Ford*, located at <http://howaboutthat.com>, accessed on 30 February, 2006.

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Marking Scheme

Name: _____

CRITERIA	CRITERIA	MARKS	MARKS GIVEN
Type of safety feature	Not mentioned	0	$\frac{\quad}{2}$
	Classifies feature.	1	
	Classifies feature with reason for classification.	2	
Development of safety feature	No description.	0	$\frac{\quad}{4}$
	Brief description of development, mentions inventor.	1 - 2	
	Good description of development; includes recent improvements.	3 - 4	
Working of the safety feature	No description.	0	$\frac{\quad}{5}$
	Brief description of how the safety feature works.	1 - 2	
	Good description given, including how the feature protects passengers.	3 - 5	
Physics principles	Not mentioned.	0	$\frac{\quad}{4}$
	Brief description of basic Physics principles.	1 - 2	
	Good description - relates Physics principles to how the feature protects passengers.	3 - 4	
Literacy	Good spelling and grammar shown.	0 - 1	$\frac{\quad}{5}$
	Information presented in own words (generally).	0 - 2	
	Bibliography presented in correct format and shows a range of resources.	0 - 2	
		Total	$\frac{\quad}{20}$

Comment